ASSIGNMENT/ASSESSMENT ITEM COVER SHEET

Student Name: Hari Krishna (FIRST NAME) **Jadda** (FAMILY / LAST NAME)

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Course Code	Course Title
I N F T 6 8 0 0	Professional Practice in IT
(Example) A B C D 1 2 3 4	*(Example)* Intro to University

Campus of Study: CALLAGHAN (eg Callaghan, Ourimbah, Port Macquarie)

Assessment Item Title: Individual Reflective Assessment Due Date/Time: 08/08/2025

Tutorial Group (If applicable): Word Count (If applicable):

Lecturer/Tutor Name: Prof. Geoffrey Skinner

Extension Granted: ○ Yes ○ No Granted Until:

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THE UNIVERSITY OF NEWCASTLE AUSTRALIA

INFT6800 – Professional Practice in IT

Individual Reflective Assessment

Name: Hari Krishna Jadda

Student ID: c3511138

Course Coordinator: Prof. Geoffrey Skinner

Lecture Timings: Tuesday 12pm

Lab Timings: Thursday 10am – 12pm

Introduction:

This assignment meets the course requirements through a detailed industry skills research, taking stock of personal skills, maintaining regular reflective journal entries, writing an academic reflection, and developing competency based interview questions. It is intended that these steps will close the gap from where I am currently at now to where I am required to be in the IT profession both in terms of technical capabilities and professional development. This assignment will provide an opportunity for critical reflection on job advertisements and to think carefully about my learning experiences, while outlining my development using the STAR method. The assignment will serve as my career map to make sure I am prepared for working in the dynamic and evolving world of technology.

Step 1: Industry Skill Research

Below are the 5 job postings that are considered for skill analysis. The skills and qualifications from these roles are relevant to my career goals for a Full Stack Developer and are sourced from LinkedIn and SEEK.

1. https://www.linkedin.com/jobs/collections/recommended/?currentJobId=4230205529 – Software Engineer Intern (Posted on 27th May, 2025)
2. https://www.linkedin.com/jobs/collections/recommended/?currentJobId=4232859648 – Java Full Stack Engineer (Posted on 27th May, 2025)

3. https://www.linkedin.com/jobs/collections/recommended/?currentJobId=4226064001&discover=recommended&discoveryOrigin=JOBS_HOME_JYMBII – Java Engineer (Posted on 21th May, 2025)
4. https://www.seek.com.au/job/84471383?ref=recomhomepage&pos=28#sol=f4508b93f7c06219f2e81684249653c9247db139 – Senior Software Engineer (Posted on 29th May, 2025)
5. https://www.seek.com.au/job/84301810?ref=recomhomepage&pos=2#sol=402d412f59d59693d20a4c26ee880c159d66055a – Website Developer (Posted on 25th May, 2025)

Taking a closer look at the advertised positions and descriptions, I found the most common technical and non-technical skills employers are asking for. This list summarizes the most common skills that appear in many jobs.

Skill Category	Skills Required
Technical Skills	Java, Spring Boot, REST APIs, Angular, HTML/CSS, Python (Flask), JavaScript, Microservices, Kafka, Docker, AWS, Git, SQL/NoSQL (MySQL, MongoDB, Cassandra), CI/CD with Jenkins or GitHub Actions
Non-Technical Skills	Communication, Problem-solving, Agile/Scrum methodology, Time Management, Team Collaboration, Adaptability

Step 2: Personal Skills Audit

As a Full Stack Developer, working for over 5 years with different companies and technologies, I have built strong abilities in both technical and non-technical areas. Currently pursuing a Master of Information Technology at the University of Newcastle, I'm actively enhancing my skills to keep up with the evolving technology industry.

Below is a reflective assessment of my current skillset with respect to the industry requirements sourced in Step 1:

Technical Strengths:

I have strong expertise in Java (Spring Boot), Python (Flask), relational and nonrelational databases like MySQL, MongoDB and Cassandra and CI/CD platforms such as Jenkins and GitHub Actions. I have led web development project projects and integrated real-time services using Kafka and RESTful APIs. In addition, I possess moderate experience with Angular and built front-end components using Angular, HTML/CSS, JavaScript and integrated them with backend services.

Non-Technical Strengths:

My strengths in non-technical skills involve problem-solving, communication, teamwork and debugging complex code. I am good at managing time effectively to achieve deadlines and can collaborate with cross-functional teams while representing the company to clients.

Areas of Development:

While I have significant expertise in full stack development, I recognize several key areas for growth to enhance competitiveness. I am currently focused on increasing my knowledge in Cloud and DevOps platforms like AWS, Azure, Kubernetes to meet the increased needs in the industry. I am also planning to develop skills in integrating AI/ML into the applications and present better versions of myself with enhanced communication, leadership and project management skills for future senior or architect jobs. I believe my postgraduate studies will fill the gaps in the development of these skills, most importantly cloud computing, DevOps and advanced software architecture. Additionally, I'm pursuing self-directed learning via online platforms to build a well-rounded skill set aligned with the changing needs of the tech industry.

Step 3: Ongoing Reflective Journal Entries

Journal Entry 1: Professional Ethics and Responsibility (Week 2-4)

Reflection Component	Description
Focus Event	Exposure to professional ethics frameworks and legal responsibilities through course lectures
Context	INFT6800 lectures on professional codes of conduct, legal frameworks, and ethical decision-making

REFLECTIVE ASSESSMENT

Reflection Component	Description
Key Learning	Understanding the ACS Code of Ethics and its practical application in IT scenarios

WHAT? During weeks 2-4 of INFT6800, I engaged with content on professional ethics, legal responsibilities, and certification frameworks. The lectures by Dr. Skinner highlighted the Australian Computer Society (ACS) Code of Ethics, professional standards, and the legal implications of IT work including intellectual property, contract agreements, and occupational health and safety requirements.

SO WHAT? This experience changed my perspective on professional practice beyond technical skills. I had previously focused the idea of professional development more on technical skills like programming, frameworks or tools, and systems design. But, the ethics content surfaced the significance of professional responsibility, ethical decision-making, and legal obligation in information technology roles. Before I hadn't thought about intellectual property protection, non-disclosure agreements, or professional liability, but when develop those understandings, I've begun to realize that technical skills must be balanced by ethical reasoning and legal awareness.

As an individual whose career goals include IT, the discussion on certification really hit home with me. Talking about stuff like professional certifications, such as ACS membership and industry specific certifications (AWS, Azure, Cisco) really opened my eyes then in terms of career aspirations. As a result, I will be pursuing some certifications that match up with things I am studying and which align with my aspirations.

NOW WHAT? Moving forward, I will integrate ethical considerations into my technical decision-making processes. I plan to pursue ACS membership upon graduation and maintain awareness of legal requirements in my development work. This reflection has also reinforced my commitment to continuous professional development through certification pathways, particularly in cloud technologies where industry credentials carry significant weight.

Journal Entry 2: AI Integration and Future Skills (Week 6-8)

Reflection Component	Description
Focus Event	Group project research on AI in professional environments and personal AI exposure
Context	Research seminar preparation and analysis of AI impact on IT careers
Key Learning	Understanding AI as augmentation rather than replacement for IT professionals

WHAT? In the middle trimester, our team project was to research AI impacting professional workplace environments. By interpreting McKinsey reports and academic writings, we researched how AI is shifting workplace practices across multiple sectors. At the same time as we were doing this, the Week 6 lecture "Ethics and AI" was giving us a theoretical analysis of AI in society.

SO WHAT? This study influenced my perspective on how AI relates to my future career in a fundamental way, although I had a minor amount of fear about AI and the prospect of a loss of jobs in IT setting. But our findings determined that AI is a tool, as a "superagency" tool, that extends, but does not replace, human capacity-in other words, we're extended by it. In sum for the IT professional, AI is a method of focusing on interesting work, or creative problem-solving, system design and architecture, and client relationships, rather than low level coding and testing like telling AI what to do in an automated manner.

The research process also highlighted the importance of developing "AI-plus" skills combining technical expertise with AI literacy and uniquely human capabilities like ethical judgment and creative thinking. This concept has become central to my professional development strategy, recognizing that future success requires both technical proficiency and the ability to work collaboratively with AI tools.

NOW WHAT? I’m now doing my best to integrate AI tools into my development workflow to gain experience and expertise. This includes using AI driven coding tools, AI driven test automation tools, and AI driven documentation generators. I have plans to develop formal AI literacy through additional courses and certification programs. Most importantly, I am working on strengthening explicitly human skills, including creative problem solving, ethical reasoning, and stakeholder communications, that will still be of value in an AI-enhanced workplace.

Journal Entry 3: Team Collaboration & Leadership Development (Week 10-12)

Reflection Component	Description
Focus Event	Group project execution and leadership challenges during final weeks
Context	Coordinating research, writing, and presentation tasks across diverse team members
Key Learning	Practical application of organizational communication and leader responsibilities

WHAT? The last few weeks of the Trimester were a bit of an intense period of collaboration around our group project related to AI in professional contexts. I was a team member and had a coordinating role, in terms of allocating tasks, making sure we adhered to a quality standard, and managing collaboration in a group of individuals with various experiences and working styles. This involved scheduling meetings regularly, managing version control for our collaborative writing and making sure everyone's voice was heard in terms of making decisions.

SO WHAT? This experience has also provided practical insight into the organizational communication and leadership concepts discussed in the Week 7 lectures. Taking on a leadership role for a diverse team of volunteers requiring supervision and management has underscored the complexity of effective communication, in balancing a directive style of leadership with an inclusive style of leadership to encourage team collaboration and collaboration on decision making, managing disagreements regarding research direction, and overseeing equitable distribution of responsibilities and tasks. My experience leads me to believe that technical leadership is supported and sustained by the interpersonal skills and emotional intelligence of leader as opposed to just technical knowledge in the discipline.

This experience also reaffirmed the concepts of cultural awareness and people with diverse cultural backgrounds and inclusive practices discussed in the context of Week 9 in regard to accessibility and culture. We had a representation of culturally diverse people from different backgrounds and different levels of the English language. Integrating cultures, perspective and viewpoints required patience, good listening and the ability to change the way communication happened.

NOW WHAT? This has helped to confirm my interest in technical leadership roles that incorporate technical skills with the management of a team. I understand that I

must also continue to develop my interpersonal skills and leadership alongside my technical skills. I expect to find opportunities for both formal leadership training and mentoring experiences. I will also make a conscious effort to practice inclusive communication techniques and cultural sensitivity to future teams, as I believe these are foundational professional skills in Australia's multicultural IT context.

Step 4: Academic Reflection

Throughout the INFT6800 trimester, my participation in the group project on "AI in the Professional Working Environment" and engagement with course content has catalysed significant personal and professional development. This reflection identifies three key skills that have emerged or been substantially enhanced through my learning journey. They are collaborative research and analysis, ethical reasoning and professional judgment, and adaptive communication and leadership.

Skill 1: Collaborative Research and Analysis

The group project involved pulling together complex content from several academic and industry sources, including McKinsey reports, World Economic Forum publications, and peer-reviewed research about AI integration into the workplace. I began researching with the impression that this was an independent process that began with collecting individual research and then we would integrate the separate projects into our group project. This was naive to think that we could continue to complete this project as individuals, as the collaborative investigation required a more sophisticated approach.

Through working with members of diverse backgrounds, I learned how to collaboratively research. Together, we shared sources as we found them, built on each other's work, and synthesized analysis instead of just summing our individual analyses. Collaboratively researching was more robust than researching individually, producing knowledge that we would have never produced alone. For example, we collaborated on an analysis of AI's impact across industries based on individual perspectives of IT, finance, and marketing careers.

This skill directly aligns with modern IT workplace practices, where complex projects require cross functional collaboration and shared knowledge creation. The ability to conduct collaborative research and analysis will be essential in my future role as a Full Stack Developer, where understanding user requirements, market trends, and technological capabilities requires integrated team investigation rather than individual technical analysis.

Skill 2: Ethical Reasoning and Professional Judgment

The course's emphasis on professional ethics, legal responsibilities, and AI ethics significantly developed my capacity for ethical reasoning in technical contexts. Through engaging with the Australian Computer Society Code of Ethics, legal frameworks for IT practitioners, and ethical implications of AI technology, I have developed structured approaches to ethical decision-making that extend beyond technical considerations.

Working on the group project looking at the implications of AI on society made me think through the larger implications of technology decisions. Working through how AI bias, job loss, and privacy issues operatively, made me mentally move from technical possibilities to looking at social responsibilities and ethical implications. My thinking developed on this theme through lectures on professional codes of conduct and legal responsibilities that offered boundaries for ethical reasoning in a professional setting.

This improved ethical reasoning ability is vital to my professional goals as an IT leader. Technical decision-making has more ethical implications now than it ever has from data privacy, algorithmic bias, accessibility, and environmental sustainability, the issues continue to be complex. I know that if I can see ethical considerations as a simultaneous discussion to technical problem solving, I will begin to set myself apart from my peers as they will have to confront moral and legal implications of their projects in technology development without support from a responsible professional.

Skill 3: Adaptive Communication and Leadership

Through navigating collaborative processes while working on the group project, I learned a lot about my ability to adapt communication practices and lead through teamwork and collaboration. As I initially communicated with the group through my specific technical training with communicating and managing tasks and deadlines, I realized that team members from different cultures and disciplines, and varying communication styles, required a more complex approach to interpersonal skills.

I learned to adapt communication approaches based on audience needs providing detailed technical explanations for some team members while offering high-level summaries for others, facilitating inclusive discussion where all perspectives were valued, and managing conflicts constructively when team members disagreed on research direction or conclusions. This development was supported by course content on organizational communication and cultural considerations in professional practice.

The leadership dimensions of this skill were focused through taking a lead in coordinating team activities while utilizing collaborative decision making. Instead of directing team activities as the leader, I learned to enable my team to problem solve

collectively and ensure that all members contributed to the process. This process was effective in generating better quality outputs and contributed to team cohesion and their recognition of ownership on the outcomes.

This adaptive communication and leadership skill is essential for my career progression in IT, where technical roles increasingly require client interaction, cross functional collaboration, and team leadership responsibilities. The ability to communicate effectively across diverse audiences and facilitate collaborative problem solving will be crucial as I advance toward senior technical and management positions.

Integration and Future Application

These three skills collaborative research and analysis, ethical reasoning and professional judgment, and adaptive communication and leadership represent an integrated development that enhances both my technical capabilities and professional effectiveness. They reflect the course emphasis on professional practice that extends beyond technical competency to encompass ethical responsibility, collaborative capability, and leadership potential.

Moving forward, I will continue developing these skills through seeking collaborative project opportunities, engaging with professional ethics forums and continuing education, and pursuing formal leadership development programs. These capabilities position me not just as a technically competent Full Stack Developer, but as a professional capable of contributing to ethical, inclusive, and effective IT organizations.

Step 5: Competency-Based Interview Questions

Based on the three key skills identified in my academic reflection, I have developed competency-based interview questions following the STAR method (Situation, Task, Action, Result) framework as recommended by Hays Australia career guidance.

Question 1: Collaborative Research and Analysis

Interview Question**:** "Describe a time when you had to collaborate with a diverse team to research and analyse complex information. How did you ensure all team members contributed effectively to the final outcome?"

Situation**:** During my group project, while examining AI's impact on professional working environments, our team consisted of students from different cultural backgrounds and academic disciplines, with varying levels of research experience and English language proficiency.

Task: We needed to synthesize complex information from multiple academic and industry sources, including McKinsey reports and World Economic Forum publications, to produce a comprehensive analysis of AI's workplace transformation across different professional domains within a strict deadline.

Action: I started a collaborative research approach by creating shared digital workspaces in Google Docs, and communication tools like Slack to have real time access. Instead of splitting up research tasks as separate individuals, I scheduled weekly brainstorming sessions where we could collaboratively look at resources and develop integrated results. I adapted my communication style to suit the different team members, providing lengthy technical explanations for some, and providing visual notes for others. When language barriers were present, I facilitated peer translations and made sure to collect each perspective on our analysis. I also set up a peer review process with each team member's contributions being validation of other team members before submitting to the final document.

Result: Working jointly provided some insights that none of us would have achieved alone. More fundamentally, the process created strong working relationships in our research team and collectively enhanced everyone's research skills and capabilities. For these reasons, I have adopted the methodology we created as my standard approach to collaborative projects, and I have shared it with other students who are going through similar processes.

Question 2: Ethical Reasoning and Professional Judgment

Interview Question: "Tell me about a situation where you had to consider ethical implications alongside technical requirements when making a professional decision. How did you balance these competing priorities?"

Situation: During research on AI applications for our group report, we encountered concerns about algorithmic bias and job loss that could influence our recommendations. At the same time, we also found considerable evidence of AI's productivity gains and innovation advantages. This developed into a tension of presenting AI in a positive light for business adoption versus taking seriously the ethical concerns.

Task: I needed to ensure our analysis provided balanced, ethically responsible recommendations while maintaining academic rigor and practical relevance for professional audiences. The challenge was avoiding uncritical AI advocacy while providing actionable guidance.

Action: After learning about the Australian Computer Society Code of Ethics framework in INFT6800, I applied it to organize our ethical analysis. In addition, I read widely on AI ethics from academic and professional sources. I led the whole team using some structured ethical reasoning approaches to assist us in recognizing multiple stakeholder views to employees, employers, and society. I often pushed for us to develop specific recommendations for a responsible AI usage, rather than just

saying what is wrong. We even reached out to some contacts we had in industry through LinkedIn for their feedback on our ethical considerations.

Result: Our final project report also had a unique section dedicated to thinking about ethical issues, balancing our excitement for the opportunities presented by AI with an understanding of the risks and responsibilities. The ethical framework we constructed was praised directly from our course coordinator and was asked for by other student groups interested in using it for their projects. However, the most impactful outcome of the course was that this experience placed ethical reasoning as an essential element in my professional decision-making. Now I also apply similar frameworks regularly in considering technology choices and career frameworks, ensuring my technical decisions include my broader professional and social responsibilities.

Question 3: Adaptive Communication and Leadership

Interview Question: "Describe a time when you had to lead a team through a challenging project while ensuring all team members felt valued and contributed effectively. What was your approach to managing diverse working styles and maintaining team cohesion?"

Situation: As we navigated our group project, tensions developed in our team based on our differences in working style and allocation of group work. Some team members preferred detailed planning like a project manager while other members preferred to work spontaneously. In terms of communication, some group members adopted a relatively hierarchal view of authority, so when given feedback they sometimes saw me as someone who needed to be respected more than viewed as a person, this cause some disputes in my group.

Task: I needed to maintain project momentum and quality standards while ensuring all team members felt valued and contributed meaningfully to our shared goals. The challenge was adapting my leadership approach to accommodate diverse preferences while maintaining accountability and meeting our academic requirements.

Action: I initiated individual conversations with all the team members for better understand how they preferred to work, how they preferred to communicate, and if they had any concerns about team dynamics. From there, I created flexible types of meeting formats including structured formal agenda items, and open discussions for agendas that had no structure. I also developed several communication channels, formal meetings to make decisions, informal chat groups to ask and answer quick questions, and one-on-one check-ins to engage in sensitive issues. When conflicts arose, I facilitated structured discussions using techniques learned in organizational communication lectures, ensuring all voices were heard before reaching consensus. I also adapted my leadership style situationally being more directive when teams

needed clear guidance and more collaborative during creative problem-solving phases.

Result: Team cohesion improved significantly, with all members actively contributing to our final deliverables. Our presentation received positive feedback for demonstrating integrated teamwork rather than individual contributions joined together. My teammates expressed thanks specifically for fostering an inclusive environment where they felt comfortable to provide their best work. This process showed me that being a good leader requires ongoing modifications and actions, following the needs and requirements of the team and the situation. Since this project, I have used adaptive communication strategies in other collaborative contexts with consistently improved team outcomes and elevated participant satisfaction.

Conclusion:

The reflective assignment has been a significant experience for me, allowing me to align my personal and professional development with the expectations of the IT industry. The research I conducted on industry skills allowed me to identify both technical and non-technical skills that I found relevant to a Full Stack Developer role and also allowed me to establish a benchmark for my development. The skills audit focused on my personal skills, where I realised my strengths include Java, Spring Boot and working collaboratively to solve problems, and I know I would like to work further on cloud technologies and integrating AI. The reflective journal entries and academic reflective assignment were useful in building my knowledge of ethical reasoning, collaborative research, and adaptive leadership, necessary skills as I transition my career. The competency based interview questions I constructed, using the STAR method, have helped me provide real-life examples on the practical applications of these skills, and prepare for future professional working opportunities. This process has enhanced my technical skills and professional development while reinforcing my commitment to lifelong learning and conducting my work in IT ethically. Moving forwards, I look to continue to take advantage of professional development through certifications, team collaboration, and developing further leadership skills, to ensure I am a well rounded IT professional aiding with innovative and inclusive technology solutions.

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